MULTI-CURRENCY TRAVEL BUDGET ANALYSIS

Python Programming Case Study

1. Problem Statement

Travellers often spend money in multiple currencies during a trip. Manually converting expenses and including exchange fees can cause errors and make it difficult to track the budget. This program converts expenses into one base currency, adds conversion fees, compares spending with the planned budget, and identifies the largest expense category.

2. Objective

To create a Python program that accepts a travel budget, expense details, supplied exchange rates, and conversion-fee percentage. It converts each expense into the selected base currency, calculates fees and total spending, checks the budget, and displays the category with the highest expense.

3. Assumptions

- The user selects one base currency, such as INR, USD, or EUR.
- Each expense has a description, category, amount, and original currency.
- Supplied exchange rates represent base-currency units for one unit of foreign currency.
- A percentage conversion fee is applied to the converted amount.
- Amounts, rates, budget, and fee percentage are non-negative; exchange rates must be greater than zero.
- Total spending includes fees. All categories tied for the highest total are displayed.

4. Input and Output

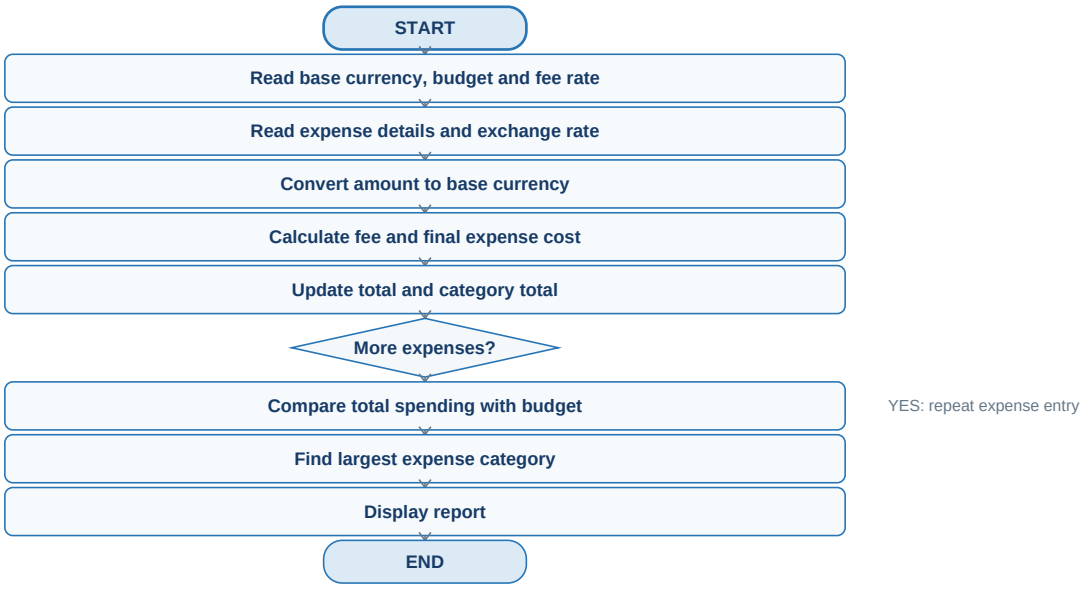
Input: Base currency, planned budget, expense count, description, category, amount, original currency, supplied exchange rate, and fee percentage.

Output: Converted amount, conversion fee, final cost per expense, total spending, budget balance or overspending, and largest expense category.

5. Algorithm

1. Start the program.
2. Read the base currency, planned budget, and conversion-fee percentage.
3. Read the number of expenses; initialize total spending and category totals to zero.
4. For each expense, read its description, category, amount, and currency.
5. Use rate 1 if the expense is already in the base currency; otherwise read the supplied exchange rate.
6. Calculate converted amount = original amount × exchange rate.
7. Calculate fee = converted amount × fee percentage ÷ 100.
8. Calculate final cost = converted amount + fee.
9. Add final cost to total spending and its category total.
10. After all expenses, compare total spending with the planned budget.
11. Find the category or categories with the highest total.
12. Display the expense report, total, budget result, and largest category.

6. Flowchart



7. Pseudocode

```
BEGIN
  INPUT base_currency, budget, fee_percent
  INPUT number_of_expenses
  total_spending = 0
  category_totals = empty dictionary
  FOR each expense:
    INPUT description, category, amount, currency
    IF currency == base_currency: rate = 1
    ELSE: INPUT supplied_exchange_rate; rate = supplied_exchange_rate
    converted_amount = amount * rate
    conversion_fee = converted_amount * fee_percent / 100
    final_cost = converted_amount + conversion_fee
    total_spending += final_cost
    category_totals[category] += final_cost
  END FOR
  IF total_spending <= budget: DISPLAY budget - total_spending
  ELSE: DISPLAY total_spending - budget as overspending
  largest_total = maximum category total
  DISPLAY total_spending and largest category/categories
END
```

8. Calculation Formulae

- Converted amount = Original amount × Exchange rate
- Conversion fee = Converted amount × Fee percentage ÷ 100
- Final expense cost = Converted amount + Conversion fee
- Budget balance = Budget – Total spending, if within budget
- Overspending = Total spending – Budget, if over budget

9. Sample Calculation

Expense	Original	Rate	Converted	Fee 2%	Final cost
Hotel (USD)	$200	■83.00	■16,600.00	■332.00	■16,932.00
Food (EUR)	€80	■90.00	■7,200.00	■144.00	■7,344.00
Local travel (INR)	■2,000	1.00	■2,000.00	■40.00	■2,040.00

Example result: Total spending = ■26,316.00. With a planned budget of ■30,000, the trip is within budget by ■3,684.00. The largest category is Hotel at ■16,932.00. Exchange rates and fee percentage are illustrative; the program uses values supplied by the user.

10. Expected Result

The program displays each expense converted into the selected base currency, includes conversion fees in total spending, identifies whether the trip is within or over budget, and reports the largest expense category. This helps travellers track expenses accurately across multiple currencies.